BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

CHEMISTRY — SSC-I (9th Grade) — ANSWER KEY (Version 1)

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

1. The subatomic particle that has no charge is:

Answer: (C) Neutron 1 mark

The **neutron** is electrically neutral with a relative charge of **0** and a mass of approximately 1 amu (1.6749×10^{-27} kg). Neutrons are present in the nucleus along with protons and are held together by the strong nuclear force.

2. The atomic number of an element is equal to the number of:

Answer: (B) Protons 1 mark

The **atomic number (Z)** of an element is defined as the number of **protons** in the nucleus of its atom. It is unique for each element and determines its identity and position in the periodic table. In a neutral atom, the number of protons equals the number of electrons.

3. The number of neutrons in $^{37}_{17}Cl$ is:

Answer: (C) 20 1 mark

Number of neutrons = Mass number (A) − Atomic number (Z) = 37 − 17 = **20**. Chlorine has two natural isotopes: Cl-35 (75.77%) with 18 neutrons and Cl-37 (24.23%) with 20 neutrons.

4. The maximum number of electrons that a *p* sub-shell can accommodate is:

Answer: (B) 6 1 mark

The **p** sub-shell can accommodate a maximum of **6 electrons**. The maximum capacity of each sub-shell is: **s = 2, p = 6, d = 10, f = 14**.

5. Group 1 elements of the periodic table are called:

Answer: (C) Alkali metals 1 mark

Group 1 elements (Li, Na, K, Rb, Cs, Fr) are called **alkali metals**. They have one valence electron in their s-sub-shell, are highly reactive, soft, and readily lose their valence electron to form +1 cations.

6. The total number of periods in the modern periodic table is:

Answer: (C) 7 1 mark

The modern periodic table has **7 horizontal rows called periods**. The first three are short periods, and the remaining four are long periods. The number of elements per period ranges from 2 (Period 1) to 32 (Period 6).

7. The element with the highest electronegativity in the periodic table is:

Answer: (B) Fluorine 1 mark

Fluorine has the highest electronegativity value (4.0 on the Pauling scale). Electronegativity is the ability of an atom to attract shared electrons toward itself in a chemical bond. It generally increases across a period and decreases down a group.

8. As we move from left to right across a period, the atomic size:

Answer: (B) Decreases 1 mark

Atomic size **decreases** across a period because the nuclear charge increases while electrons are added to the same valence shell. The stronger nuclear pull on the valence electrons reduces the radius of the atom.

9. A chemical bond formed by the complete transfer of electrons is called:

Answer: (B) Ionic bond 1 mark

An **ionic bond** is a strong electrostatic attraction formed between oppositely charged ions when one atom (usually a metal) loses electrons to form a cation, and another atom (usually a non-metal) gains those electrons to form an anion.

10. The number of shared pairs of electrons in a nitrogen molecule (N_2) is:

Answer: (C) Three 1 mark

Nitrogen (Group 15) has 5 valence electrons and needs 3 more to complete its octet. Two nitrogen atoms share **three pairs of electrons** forming a **triple covalent bond** (N≡N).

11. Which of the following compounds contains a coordinate covalent bond?

Answer: (C) NH_4^+ 1 mark

In the **ammonium ion (NH_4^+)**, the nitrogen atom donates its lone pair of electrons to a proton (H^+) from HCl, forming a **coordinate covalent bond**. In a coordinate bond, both shared electrons come from a single atom (the donor).

12. Hydrogen bonding occurs when hydrogen is bonded to:

Answer: (B) N, O, F 1 mark

Hydrogen bonding occurs when a hydrogen atom is covalently bonded to a highly electronegative atom such as **nitrogen (N), oxygen (O), or fluorine (F)**. The electron-deficient hydrogen is then attracted to a lone pair on another electronegative atom of a nearby molecule.

SECTION B — Short Questions (11 × 3 = 33 Marks)

Q.2(i): State the main postulates of Dalton's atomic theory. Why is it considered the foundation of modern chemistry? 3 marks

Dalton's Atomic Theory (1803): John Dalton proposed the following postulates: 2 marks

1. All elements are composed of tiny indivisible particles called **atoms**.
2. Atoms of a particular element are **identical** — they have the same mass and same volume.
3. During chemical reactions, atoms **combine, separate, or rearrange** in simple ratios.
4. Atoms can **neither be created nor destroyed**.

Importance: Dalton's theory explained the laws of chemical combinations and provided a quantitative basis for chemistry. It is considered the foundation of modern chemistry because it introduced the concept of the atom as the building block of matter. 1 mark

OR

Q.2(i) OR: State any three defects in Rutherford's atomic model. 3 marks

Rutherford's planetary model resembles our solar system but has the following defects: 1 mark each = 3 marks

1. According to **classical physics**, an electron being a charged particle would emit energy continuously while revolving around the nucleus. As a result, the orbit of the revolving electron would become smaller and smaller until the electron falls into the nucleus, collapsing the atomic structure.
2. If the revolving electron emits energy continuously, it should produce a **continuous spectrum**. However, the actual spectrum of hydrogen is a **line spectrum**, not continuous.
3. Rutherford's model could not explain the **stability of the atom** nor the arrangement of electrons in atoms with more than one electron.

Q.2(ii): Compare the relative charge and relative mass of proton, neutron, and electron in tabular form. 3 marks

Particle	Relative Charge	Relative Mass	Location
Proton	+1	~1 amu (1.6726×10^{-27} kg)	Nucleus
Neutron	0 (neutral)	~1 amu (1.6749×10^{-27} kg)	Nucleus
Electron	−1	~1/1836 amu (9.11×10^{-31} kg)	Shells around nucleus

Protons and neutrons together are called **nucleons** and contribute almost the entire mass of the atom. 1 mark for each row = 3 marks

OR

Q.2(ii) OR: Differentiate between atomic number (Z) and mass number (A) with one example. 3 marks

Property	Atomic Number (Z)	Mass Number (A)
Definition	Number of **protons** in the nucleus of an atom 1 mark	Total number of **protons + neutrons** in the nucleus 1 mark
Symbol	Z	A
Determines	Identity and position in periodic table	Mass of the atom
Example		

	For $^{23}_{11}Na$: Z = 11 (protons), A = 23 (11 protons + 12 neutrons) 1 mark

Q.2(iii): Define isotopes. Describe the three isotopes of hydrogen with their symbols and number of neutrons. 3 marks

Isotopes: Atoms of the same element that have the **same atomic number but different mass numbers** due to different numbers of neutrons. The word "isotope" was first used by Soddy and comes from the Greek *isos* (same) and *tope* (place). 1 mark

Three Isotopes of Hydrogen: ⅔ mark each = 2 marks

1. **Protium (^1_1H):** 1 proton, **0 neutrons**. Most abundant (99.99%).
2. **Deuterium (^2_1H or D):** 1 proton, **1 neutron**. Found in heavy water (D_2O), abundance 0.0015%.
3. **Tritium (^3_1H or T):** 1 proton, **2 neutrons**. Radioactive and very rare.

OR

Q.2(iii) OR: State any three uses of radioactive isotopes in medicine and research. 3 marks

Radioactive isotopes have many important applications: 1 mark each = 3 marks

1. **Iodine-131** is used as a tracer in **diagnosing thyroid problems**, and **Iodine-123** is used to image the brain.
2. **Cobalt-60** is used to **irradiate cancer cells** in radiotherapy to kill or shrink tumors.
3. **Carbon-14** is used in **carbon dating** to estimate the age of rocks, fossils, archaeological objects, and mummies. **Sodium-24** is used to trace the flow of blood and detect blood clots.

Q.2(iv): Differentiate between a cation and an anion with one example of each. 3 marks

Property	Cation	Anion
Definition	Positively charged ion formed when an atom **loses** one or more electrons 1 mark	Negatively charged ion formed when an atom **gains** one or more electrons 1 mark
Formed by	Metals (low ionization energy)	Non-metals (high electron affinity)
Example	$Na \rightarrow Na^+ + e^-$	$Cl + e^- \rightarrow Cl^-$
Resulting config	Both achieve stable noble gas electronic configuration 1 mark	

OR

Q.2(iv) OR: Write the electronic configuration of $_{11}Na$, $_{13}Al$, and $_{17}Cl$. 3 marks

Using the Aufbau principle, electrons fill the lowest-energy sub-shell first in the order: 1s < 2s < 2p < 3s < 3p ... 1 mark each = 3 marks

- **$_{11}$Na:** $1s^2, 2s^2, 2p^6, 3s^1$
- **$_{13}$Al:** $1s^2, 2s^2, 2p^6, 3s^2, 3p^1$
- **$_{17}$Cl:** $1s^2, 2s^2, 2p^6, 3s^2, 3p^5$

Q.2(v): Define periodic table and state the modern periodic law. 3 marks

Periodic Table: A table that shows the systematic arrangement of all known elements in **order of increasing atomic number** in horizontal rows (periods) and vertical columns (groups), so that elements with similar properties fall in the same group. 1.5 marks

Modern Periodic Law (Moseley, 1913): "If the elements are arranged in the order of their **increasing atomic numbers**, their physical and chemical properties are **repeated in a periodic manner**." 1.5 marks

OR

Q.2(v) OR: Differentiate between a group and a period in the periodic table. 3 marks

Property	Group	Period
Definition	**Vertical column** of elements 1 mark	**Horizontal row** of elements 1 mark
Number	18 groups (1 to 18 IUPAC)	7 periods
Elements share	Same number of valence electrons — **similar chemical properties**	Same outermost shell (n value); properties change gradually 1 mark

Q.2(vi): Define shielding effect. How does it vary across a period and down a group? 3 marks

Shielding Effect: The reduction in the force of attraction between the nucleus and the valence electrons, caused by the electrons present in the inner sub-shells. The inner electrons "shield" or screen the outer electrons from the full nuclear pull. 1 mark

Variation:

- **Across a period (left to right):** Shielding effect **remains constant** because the number of electrons in the inner shells does not change. [1 mark]

- **Down a group (top to bottom):** Shielding effect **increases** because the number of inner electronic shells increases. [1 mark]

OR

Q.2(vi) OR: Define ionization energy. How does it change across a period and down a group? [3 marks]

Ionization Energy: The minimum amount of energy required to **remove the outermost electron** from an isolated gaseous atom. [1 mark]

$$M_{(g)} + \text{ionization energy} \rightarrow M^+_{(g)} + e^-$$

Trends:

- **Across a period:** Ionization energy **increases** because nuclear charge increases and atomic size decreases, making it harder to remove the valence electron. [1 mark]
- **Down a group:** Ionization energy **decreases** because the atomic size and shielding effect increase, so the valence electron is more loosely held. [1 mark]

Q.2(vii): State any three characteristic properties of alkali metals (Group 1 elements). [3 marks]

Characteristic Properties of Alkali Metals: [1 mark each = 3 marks]

1. **Highly reactive metals:** They readily lose their single valence electron to form **+1 cations**. Reactivity increases down the group.
2. **Soft and low density:** They are so soft they can be cut with a knife. They have low densities (Li, Na, K float on water).
3. **Excellent conductors:** They are good conductors of heat and electricity, have low melting points, and react vigorously with water to produce metal hydroxides and hydrogen gas:
   $2Na + 2H_2O \rightarrow 2NaOH + H_2\uparrow$

OR

Q.2(vii) OR: Why are noble gases (Group 18) chemically inert? Explain in terms of electronic configuration. [3 marks]

Noble Gases (He, Ne, Ar, Kr, Xe, Rn) are found in Group 18 of the periodic table and are also called **inert gases**. [1 mark]

Reason for Inertness: Noble gases have **completely filled valence shells** with the general electronic configuration $\mathbf{ns^2np^6}$ (8 electrons), except helium which has $1s^2$ (2 electrons). This **complete octet** (or duplet for He) makes them chemically stable. [1.5 marks]

Examples:

- He = $1s^2$
- Ne = $1s^2, 2s^2, 2p^6$
- Ar = $1s^2, 2s^2, 2p^6, 3s^2, 3p^6$

Because they neither tend to lose, gain, nor share electrons, they do not form chemical bonds under ordinary conditions. [½ mark]

Q.2(viii): Compare the general physical properties of metals and non-metals (any three differences). [3 marks]

Property	Metals	Non-Metals
Thermal & Electrical Conductivity	Good conductors (free electrons can move freely in metal lattice) [1 mark]	Poor conductors (lack free electrons; graphite is an exception)
Malleability & Ductility	Malleable and ductile — can be hammered into sheets and drawn into wires [1 mark]	Brittle — neither malleable nor ductile
Melting & Boiling Points	High due to strong metallic bonds [1 mark]	Generally low (held by weak intermolecular forces)

OR

Q.2(viii) OR: Describe any three characteristic properties of transition elements. [3 marks]

Transition elements lie in the d-block (Groups 3 to 12) of the periodic table. They show the following characteristic properties: [1 mark each = 3 marks]

1. **High density and high melting points:** Due to higher atomic masses, closely packed structures, and strong metallic bonding from partially filled d-sub-shells. For example, Fe = 7.87 g/cm^3, W melts at 3422 °C.
2. **Variable oxidation states:** Because both d and s sub-shell electrons can participate in bonding. For example, Fe shows +2 and +3, Cu shows +1 and +2.
3. **Coloured compounds and catalytic activity:** Their compounds are often coloured (Cu compounds are blue/green; Cr compounds are red/green). They are widely used as catalysts — e.g., Fe in the Haber process, Ni

in the manufacture of margarine, Pt in catalytic converters.

Q.2(ix): State the octet and duplet rules. Give one example of each. 3 marks

Octet Rule: Atoms of the major group elements tend to participate in chemical bonding so that they acquire **eight electrons** in their valence shell (the configuration of the nearest noble gas). 1 mark

Example: Sodium loses one electron and chlorine gains one electron, both attaining 8 valence electrons (Na^+ → Ne config, Cl^- → Ar config). ½ mark

Duplet Rule: Atoms close to helium (H, Li, Be) tend to acquire **two electrons** in their valence shell to attain the helium configuration. 1 mark

Example: Lithium loses one electron to form Li^+ ($1s^2$); hydrogen gains one electron to form H^- ($1s^2$). ½ mark

OR

Q.2(ix) OR: Why do atoms react with each other? Explain with reference to noble gas configuration. 3 marks

Atoms react with each other to **achieve stability**. 1 mark

Noble gases (Group 18) are the most stable elements because they have **complete valence shells** (8 electrons, or 2 in helium). All other elements have incomplete valence shells, making them reactive. 1 mark

To attain a stable noble gas configuration, atoms can: 1 mark

1. **Lose electrons** — metals (electropositive) lose electrons to form cations (e.g., Na → Na^+).
2. **Gain electrons** — non-metals (electronegative) gain electrons to form anions (e.g., Cl → Cl^-).
3. **Share electrons** — non-metals share electrons with each other to form covalent bonds (e.g., H_2, O_2).

Q.2(x): Define ionic bond. Show the formation of NaCl using electron dot and cross structures. 3 marks

Ionic Bond: A strong electrostatic force of attraction between **positively charged metal ions and negatively charged non-metal ions**, formed when one atom completely transfers one or more electrons to another atom. 1 mark

Formation of NaCl: 2 marks

- Sodium (Na) belongs to Group 1 and has 1 valence electron. It loses this electron to form Na^+ ion (achieving Ne configuration).
- Chlorine (Cl) belongs to Group 17 and has 7 valence electrons. It gains one electron to form Cl^- ion (achieving Ar configuration).

```
Na ·  +  ×××××××Cl  →  [Na]⁺ [·×××××××Cl]⁻
```

The Na^+ and Cl^- ions are held together by strong electrostatic forces, forming the ionic compound **NaCl (sodium chloride)**.

OR

Q.2(x) OR: Differentiate between polar and non-polar covalent bonds with one example of each. 3 marks

Property	Non-Polar Covalent Bond	Polar Covalent Bond
Definition	Bond formed between **two identical atoms** where electrons are shared equally 1 mark	Bond formed between **two different atoms** with different electronegativities; electrons are shared unequally 1 mark
Charge distribution	Symmetrical, no partial charges	One end is δ+, the other is δ−
Example	H—H, O=O, Cl—Cl, N≡N ½ mark	H—Cl, H—O—H, NH_3, HCN ½ mark

Q.2(xi): Define coordinate covalent bond. Show the formation of ammonium ion (NH_4^+) as an example. 3 marks

Coordinate Covalent Bond: A type of covalent bond in which the **shared pair of electrons comes from a single atom** (called the donor). Once formed, it is indistinguishable from a normal covalent bond. It is also called a **dative bond**. 1.5 marks

Formation of NH_4^+: When ammonia (NH_3) reacts with hydrogen chloride (HCl), the lone pair of electrons on nitrogen is donated to the H^+ ion (proton) from HCl, forming the ammonium ion. 1.5 marks

```
H—N··H + H—Cl  →  [H—N→H]⁺ + Cl⁻
   |                   |
  H                    H
```

The fourth N—H bond is the coordinate covalent bond because both shared electrons come from nitrogen.

OR

Q.2(xi) OR: What is hydrogen bonding? Explain with the example of water molecules. 3 marks

Hydrogen Bond: A weak intermolecular attractive force between a **highly electron-deficient hydrogen atom** (covalently bonded to a strongly electronegative atom such as N, O, or F) and a **lone pair on a nearby electronegative atom**. 1.5 marks

Hydrogen Bonding in Water: In a water molecule (H_2O), oxygen is highly electronegative and pulls the shared electrons toward itself, leaving the hydrogen atoms with a partial positive charge (δ^+) and oxygen with a partial negative charge (δ^-). 1 mark

The δ^+ hydrogen of one water molecule is then weakly attracted to the lone pair on the oxygen of another water molecule, forming a hydrogen bond. This is responsible for water's unusually high boiling point, surface tension, and its role as the universal solvent. ½ mark

SECTION C — Long Questions (4 × 5 = 20 Marks)

Q.3(i): State the main postulates of Bohr's atomic theory. How does Bohr's model overcome the defects of Rutherford's model? 5 marks

In **1913 Niels Bohr** proposed an atomic model that was consistent with Rutherford's planetary model and also explained the line spectrum of hydrogen.

Main Postulates of Bohr's Atomic Theory: 3.5 marks

1. The electron in an atom revolves around the nucleus only in **certain fixed circular orbits**. Each orbit has a fixed energy and is also called an **energy level**.
2. The energy of the electron in an orbit is **proportional to its distance** from the nucleus. The farther the electron is from the nucleus, the more energy it has.
3. The electron revolves only in those orbits for which the **angular momentum is an integral multiple of h/2π**, where h is Planck's constant (6.626×10^{-34} J·s). *Angular momentum is quantized.*
4. Light is **absorbed** when an electron jumps from a lower to a higher energy orbit, and **emitted** when an electron falls from a higher to a lower energy orbit. An electron in a particular orbit does not radiate energy.
5. The energy of the light emitted or absorbed is exactly equal to the difference between the energies of the two orbits:
   $\mathbf{\Delta E = E_2 - E_1}$

How Bohr's Model Overcomes Rutherford's Defects: 1.5 marks

- Bohr proposed that electrons revolve in **fixed orbits without radiating energy**, so the atom does not collapse.
- The atom emits a **line spectrum** (not continuous) because energy is absorbed/emitted only in discrete amounts when electrons jump between fixed orbits.
- Bohr's model successfully explained the **stability of the atom** and the line spectrum of hydrogen.

OR

Q.3(i) OR: Describe Rutherford's gold foil experiment in detail. State the conclusions he drew and the defects of his planetary model. 5 marks

Rutherford's Gold Foil Experiment (1911): 2 marks

Rutherford bombarded a very thin gold foil (about **0.00004 cm** thick) with **α-particles** obtained from the disintegration of polonium. α-particles are doubly positively charged helium nuclei (He^{2+}). He observed:

- Most of the α-particles passed straight through the foil undeflected.
- A few particles were slightly deflected.
- About 1 in 1 million particles was deflected at an angle greater than 90° (some bounced back).

Conclusions of Rutherford: 1.5 marks

1. Since most of the α-particles passed undeflected, **most of the space inside an atom is empty**.
2. The deflection of a few α-particles through large angles indicates that the positive charge of the atom is concentrated in a **small, dense, positively charged region** called the **nucleus**.
3. The mass of the atom is concentrated in this nucleus, and **electrons revolve around the nucleus** in circular orbits like planets around the sun (planetary model).

Defects of Rutherford's Planetary Model: 1.5 marks

1. According to classical physics, an accelerating (revolving) charged electron should **continuously emit energy**, spiral inward, and eventually fall into the nucleus — the atom should collapse.
2. If the electron continuously emits energy, it should produce a **continuous spectrum**, but the actual hydrogen spectrum is a **line spectrum**.
3. The model could not explain the arrangement of electrons in atoms with multiple electrons.

Q.3(ii): Define electronic configuration. Explain the concept of shells, sub-shells, and the Aufbau principle with examples. 5 marks

Electronic Configuration: The arrangement of electrons in the various sub-shells of an atom is called its electronic configuration. 1 mark

Shells (Energy Levels): According to Bohr's theory, electrons revolve around the nucleus in fixed circular orbits called **shells** or **energy levels**. Each shell is described by a principal quantum number **n** (1, 2, 3, ...). 1 mark

- n = 1 → K shell
- n = 2 → L shell
- n = 3 → M shell
- n = 4 → N shell

As n increases, the energy of the shell and its distance from the nucleus also increase.

Sub-shells: Each shell is sub-divided into sub-shells designated by letters s, p, d, f. The maximum number of electrons each sub-shell can accommodate is: 1.5 marks

Sub-shell	s	p	d	f
Max electrons	2	6	10	14

- K (n=1) has only the 1s sub-shell.
- L (n=2) has 2s and 2p sub-shells.
- M (n=3) has 3s, 3p, and 3d sub-shells.
- N (n=4) has 4s, 4p, 4d, and 4f sub-shells.

Aufbau Principle: Electrons fill the **lowest-energy sub-shell first**, then the next higher one, and so on. The order of increasing energy of sub-shells is: 1.5 marks

$$1s < 2s < 2p < 3s < 3p < 4s < 3d < \ldots$$

Examples:

- $_{11}Na$: $1s^2, 2s^2, 2p^6, 3s^1$
- $_{17}Cl$: $1s^2, 2s^2, 2p^6, 3s^2, 3p^5$
- $_{18}Ar$: $1s^2, 2s^2, 2p^6, 3s^2, 3p^6$

OR

Q.3(ii) OR: Define isotopes. Describe the isotopes of hydrogen, carbon, and chlorine with their natural abundance and uses. 5 marks

Isotopes: Atoms of the same element that have the **same atomic number but different mass numbers** due to different numbers of neutrons. They are chemically alike but differ in physical properties. 1 mark

1. Isotopes of Hydrogen: 1.5 marks

- **Protium (^1_1H):** 1 proton, 0 neutron, abundance **99.99%**. Used as fuel in rockets and in the manufacture of ammonia.
- **Deuterium (^2_1H or D):** 1 proton, 1 neutron, abundance **0.0015%**. Found in heavy water (D_2O); used as a moderator in nuclear reactors.
- **Tritium (^3_1H or T):** 1 proton, 2 neutrons; **radioactive and very rare**. Used in hydrogen bombs and biological tracing.

2. Isotopes of Carbon: 1.5 marks

- **^{12}C:** 6 protons, 6 neutrons, abundance **98.8%**. Used as the standard for atomic mass.
- **^{13}C:** 6 protons, 7 neutrons, abundance **1.1%**.
- **^{14}C:** 6 protons, 8 neutrons, abundance **0.009%**; **radioactive**. Used in **carbon dating** to estimate the age of fossils, rocks, and archaeological objects.

3. Isotopes of Chlorine: 1 mark

- **^{35}Cl:** 17 protons, 18 neutrons, abundance **75.77%**.
- **^{37}Cl:** 17 protons, 20 neutrons, abundance **24.23%**.

The relative atomic mass of chlorine (35.45 amu) is the weighted average of the masses of its isotopes.

Q.3(iii): Explain the periodic trends in atomic size, ionization energy, electron affinity, and electronegativity across a period and down a group. 5 marks

The physical and chemical properties of elements show periodic variation because their electronic configurations vary periodically with atomic number. The four main periodic properties are:

1. Atomic Size: The average distance between the nucleus and the outermost electron shell. 1 mark

- **Across a period (left to right): Decreases**, because the nuclear charge increases while electrons are added to the same valence shell, pulling them closer to the nucleus.
- **Down a group (top to bottom): Increases**, because new electron shells are added with each successive element.

2. Ionization Energy: The minimum energy required to remove the outermost electron from an isolated gaseous atom. 1.5 marks

- **Across a period: Increases**, because the nuclear charge increases and atomic size decreases, holding the valence electron more tightly.
- **Down a group: Decreases**, because atomic size and shielding effect increase, so the valence electron is held

more loosely.

3. Electron Affinity: The amount of energy released when an electron is added to the valence shell of an isolated gaseous atom. 1.5 marks

- **Across a period: Increases**, due to higher nuclear charge and smaller atomic radius, which bind the added electron more tightly. Halogens have the highest electron affinities.
- **Down a group: Decreases**, due to increased shielding effect and atomic radius.

4. Electronegativity: The ability of an atom to attract shared electrons toward itself in a chemical bond. 1 mark

- **Across a period: Increases** — F has the highest value (4.0).
- **Down a group: Decreases** — Cs has one of the lowest values (0.7).

Summary: Atomic size and metallic character decrease across a period and increase down a group, while ionization energy, electron affinity, and electronegativity show the opposite trend.

OR

Q.3(iii) OR: Describe the halogens (Group 17) in detail including their appearance, electronic configuration, reactivity, displacement reactions, and thermal stability of hydrogen halides. 5 marks

Halogens are the elements of Group 17 of the periodic table (F, Cl, Br, I, At). The name "halogen" comes from the Greek *halous* (salt) and *gen* (former), meaning "salt-formers". They are reactive non-metals and exist as **diatomic molecules** (X_2). 1 mark

1. Appearance (at room temperature): 1 mark

- **Fluorine (F_2):** pale yellow gas
- **Chlorine (Cl_2):** yellow-green gas
- **Bromine (Br_2):** red-brown liquid
- **Iodine (I_2):** grey-black solid (gives violet vapours on warming)

2. Electronic Configuration: All halogens have the general configuration ns^2np^5, with 7 valence electrons. They need only 1 electron to complete their octet, so they tend to gain one electron and form **monovalent anions** (F^-, Cl^-, Br^-, I^-). 1 mark

3. Reactivity: Halogens are highly reactive non-metals. Their reactivity **decreases down the group** because the atomic size increases and electronegativity decreases. **Fluorine is the most reactive** and **iodine is the least reactive**. The order of decreasing oxidizing power is: 1 mark

$$F_2 > Cl_2 > Br_2 > I_2$$

4. Displacement Reactions: A more reactive halogen displaces a less reactive halide ion from its salt solution. For example:

$$F_{2(g)} + 2KCl_{(aq)} \rightarrow 2KF_{(aq)} + Cl_{2(g)}$$
$$Cl_{2(g)} + 2KBr_{(aq)} \rightarrow 2KCl_{(aq)} + Br_{2(l)}$$

However, I_2 cannot displace any of the other halide ions. ½ mark

5. Thermal Stability of Hydrogen Halides: Halogens react with hydrogen to form hydrogen halides (HF, HCl, HBr, HI). The H—X bond strength depends on the electronegativity difference between H and X. As we go down the group, electronegativity decreases, so the bond becomes weaker. Thermal stability decreases in the order: ½ mark

$$HF > HCl > HBr > HI$$

Q.3(iv): Explain the formation of a covalent bond. Describe single, double, and triple covalent bonds with the examples of H_2, O_2, and N_2 using electron dot and cross diagrams. 5 marks

Covalent Bond: A chemical bond formed by the **mutual sharing of electrons** between two non-metal atoms is called a covalent bond. The shared electron pair is attracted by the nuclei of both atoms, holding them together. 1 mark

A covalent bond is represented by a single line (single bond), double line (double bond), or triple line (triple bond) between the bonded atoms. Pairs of valence electrons that are not shared are called **lone pairs**.

1. Single Covalent Bond — H_2: 1.5 marks

Each hydrogen atom has 1 valence electron. To attain the helium duplet, two H atoms share **one pair of electrons**, forming a single covalent bond.

```
H· + ×H  →  H:H  or  H—H
```

2. Double Covalent Bond — O_2: 1.5 marks

Oxygen (Group 16) has 6 valence electrons and needs 2 more to complete its octet. Two oxygen atoms share **two pairs of electrons**, forming a double covalent bond.

```
··O·· + ××O××  →  ··O::O··  or  O=O
```

3. Triple Covalent Bond — N_2: 1 mark

Nitrogen (Group 15) has 5 valence electrons and needs 3 more to complete its octet. Two nitrogen atoms share **three pairs of electrons**, forming a triple covalent bond.

```
···N + N×××  →  :N≡N:  or  N≡N
```

The triple bond in N_2 is very strong, making nitrogen gas chemically inert at room temperature.

OR

Q.3(iv) OR: Define metallic bond. Explain the electron-sea model and use it to account for the malleability, ductility, and electrical conductivity of metals. 5 marks

Metallic Bond: The strong electrostatic force of attraction between the **positively charged metal ions (cations)** and the **sea of delocalized valence electrons** in a metal lattice is called a metallic bond. 1 mark

Electron-Sea Model: 1.5 marks

In a metal, the valence electrons are not bound to any particular atom. Instead, metal atoms lose their valence electrons, becoming positive ions (cations), and these released electrons are **free to move throughout the entire metal lattice**. They form a "sea" or "cloud" of **delocalized electrons** in which the metal cations are immersed.

The strong attraction between the positive metal ions and the negatively charged electron sea constitutes the metallic bond. This model successfully accounts for the characteristic properties of metals:

1. Malleability and Ductility: 1 mark

When force is applied, the layers of metal cations can **slide over one another** without breaking the metallic bond, because the electron sea continues to bind them. This allows metals to be hammered into thin sheets (malleability) and drawn into wires (ductility).

2. Electrical Conductivity: 1 mark

Metals are excellent conductors of electricity because the **delocalized electrons can move freely** throughout the metal lattice under the influence of an applied electric field, carrying charge from one point to another.

3. Other properties: ½ mark

- **Thermal conductivity:** The free electrons also transfer thermal energy efficiently from one place to another.
- **High melting and boiling points:** Due to the strong metallic bond, metals have giant structures that require a large amount of energy to break, making them thermally stable.
- **Metallic lustre:** The free electrons reflect light, giving metals their characteristic shiny appearance.